01 · LESSON

# Plant products: starch, lipids and proteins

*9B Plant growth · Lesson 4 of 7*

Date _____ Class _____

**Today I can**

- ☐ I can describe how plants use glucose to make starch, cellulose, lipids and proteins, and explain why nitrates are needed.
- ☐ I can test a leaf for starch and explain what happens when a seed germinates.

**What happens today**

1. Do Now: potato and iodine
2. Learn: Carbohydrates: starch and cellulose · Lipids and proteins · and more
3. Practical: Testing a variegated leaf for starch
4. Challenge, then the exit ticket

Student Book pages 26 to 27 · Workbook pages 19

**What I learned today (one or two sentences)**



02 · KEY VOCABULARY

# Key Vocabulary

| Term | Definition |
|---|---|
| **starch** | a polymer made of many glucose molecules joined together; a plant's energy store |
| **polymer** | a long molecule made of many small molecules joined in a chain |
| **cellulose** | a polymer of glucose used to make plant cell walls |
| **lipid** | a fat or oil; insoluble in water |
| **protein** | a polymer made of amino acids; enzymes are proteins |
| **amino acid** | a small molecule that proteins are built from; needs nitrogen |
| **enzyme** | a protein that speeds up a reaction in a living thing |
| **nitrate** | a mineral salt that supplies nitrogen for making amino acids |
| **germination** | when a seed starts to grow into a seedling |
| **iodine solution** | an orange-brown liquid that turns blue-black with starch |
| **Benedict's solution** | a blue liquid that turns orange-red when heated with glucose |



03 · QUICK REVIEW

# Quick Review

*potato and iodine*

**1** What colour change would you see if you added iodine solution to a slice of potato?

**2** Which substance in the potato causes this change?

**3** Which substance made in photosynthesis was used to make it?



04 · CORE

# Core 1 of 4

*Everyone does these. Answer on the card or under it.*

**1** Practical: testing a variegated leaf for starch.

**a)** Draw the leaf before the test and after adding iodine. Label the green and white parts on the first drawing, and the blue-black parts on the second.

| Before the test | After adding iodine |
|---|---|
| | |

# Core 2 of 4

*Everyone does these. Answer on the card or under it.*

**1** *continued*

**b)** Name the solution used to test for starch and the colour it turns when starch is present.

**c)** Explain why only some parts of the leaf changed colour.





# Core 3 of 4

*Everyone does these. Answer on the card or under it.*

**2** Complete the sentences. Use the word bank.

Glucose molecules made in photosynthesis are joined together to make a polymer called _____.

Starch is an energy store. It is found in the chloroplasts, in storage organs such as _____, and in seeds.

Sugars are also used to make _____, which builds plant cell walls, and _____ such as the oils in seeds.





# Core 4 of 4

*Everyone does these. Answer on the card or under it.*

**3** Match each substance to one of its uses in a plant. Draw a line.

| | |
|---|---|
| starch | speed up reactions such as photosynthesis and respiration |
| cellulose | energy store in seeds and potatoes |
| lipids | build cell walls |
| proteins | built from amino acids using nitrogen from nitrates |
| enzymes | make the waterproof cuticle and cell membranes |





# Challenge 1 of 2

*When Core is finished. Full sentences.*

**C1** A plant was left in the dark for 24 hours. A metal foil stencil was then attached to one leaf and the plant was left in bright light for a day. The leaf was then tested for starch.

**a)** Explain why the plant was left in the dark first.

**b)** Predict the result of the starch test and explain it.

# Challenge 2 of 2

*When Core is finished. Full sentences.*

**C1** *continued*

**c)** Explain why a metal foil stencil was used and not a clear plastic one.

**C2** Avocado fruit flesh is rich in lipids. Avocado seeds also contain lipids. Explain why the plant puts lipids in each place. The two reasons are different.





# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** Which solution tests for starch, and what colour does it turn?

**2** Why do seeds need a store of starch?

**3** What are enzymes made of, and which mineral salt do plants need to make them?

**Today I can:** Goal 1 ☐ Goal 2 ☐